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Github: https://github.com/zere0802/rwa-tokenization-platform

SECURITY AUDIT REPORT: RWA Tokenization Platform

Executive Summary

This report presents the results of an internal security review conducted for the RWA Tokenization Platform project developed as part of the Blockchain Technologies 2 Final Project.

The protocol includes:

- ERC20 governance token
- ERC4626 vault
- DAO governance
- Chainlink oracle integration
- CREATE2 deployment factory
- Security testing suite
- Fuzz and invariant testing

The audit focused on:

- Access control security
- Governance security
- Reentrancy vulnerabilities
- Oracle manipulation risks
- Upgradeability risks
- Treasury protection
- Smart contract correctness

The assessment included:

- Manual review
- Slither static analysis
- Unit testing
- Fuzz testing
- Invariant testing
- Fork testing

No Critical, High, or Medium severity vulnerabilities remained unresolved in the final reviewed version.

Scope

Repository: RWA Tokenization Platform

In-Scope Contracts

- contracts/token/RWAToken.sol
- contracts/vault/RWAVault.sol
- contracts/governance/Governance.sol
- contracts/oracle/OracleAdapter.sol
- contracts/factory/RWAFactory.sol
- contracts/utils/YulMath.sol
- contracts/mocks/SecureVault.sol
- contracts/mocks/SecureTreasury.sol

Out-of-Scope

- Frontend code
- Wallet providers
- External protocols
- Future upgradeable contracts

Methodology: The audit combined automated tooling with manual review.

Static Analysis: The project was analyzed using Slither.

Checks included:

- Reentrancy risks
- Uninitialized storage
- Access-control issues
- Dangerous external calls
- Arithmetic issues
- Dead code
- Naming inconsistencies

Manual Review

Manual review focused on:

- Governance lifecycle correctness
- Oracle staleness validation
- Vault accounting
- ERC4626 compliance
- Access-control enforcement
- Timelock configuration
- Treasury ownership assumptions

Dynamic Testing

The protocol includes:

- Unit tests
- Fuzz tests
- Invariant tests
- Fork tests

Coverage exceeded 90% across contracts/.

Findings Summary

ID	Severity	Title	Status
S-01	Low	Oracle stale price risk	Fixed
S-02	Low	Reentrancy vulnerability example	Fixed
S-03	Low	Missing access restriction example	Fixed
S-04	Informational	Centralized governance bootstrap	Acknowledged
S-05	Informational	Governance whale voting power	Acknowledged

S-06	Gas	Repeated storage reads	Acknowledged

Detailed Findings

S-01 — Oracle Stale Price Risk

Severity: Low
Location: contracts/oracle/OracleAdapter.sol

Description

If oracle data is not validated for freshness, stale prices may be used by downstream protocol logic.

Impact

Stale prices could cause incorrect valuation and inaccurate protocol behavior.

Recommendation

Implement staleness checks using timestamps.

Resolution

The OracleAdapter contract validates that prices are not older than the configured threshold.

Status

Fixed.

S-02 — Reentrancy Vulnerability Example

Severity: Low
Location: contracts/mocks/VulnerableVault.sol

Description

A deliberately vulnerable vault implementation demonstrated how reentrancy attacks can occur when state updates happen after external calls.

Impact

An attacker may repeatedly withdraw funds before balances are updated.

Recommendation

Use the Checks-Effects-Interactions pattern or ReentrancyGuard.

Resolution

A secure implementation was added using protected state update ordering.

Status

Fixed.

S-03 — Missing Access Restriction Example

Severity: Low
Location: contracts/mocks/VulnerableTreasury.sol

Description

A deliberately insecure treasury implementation demonstrated how unrestricted administrative actions may expose protocol funds.

Impact

Unauthorized users may withdraw treasury assets.

Recommendation

Protect privileged functionality using AccessControl.

Resolution

SecureTreasury.sol introduced proper role-based permissions.

Status

Fixed.

S-04 — Centralized Governance Bootstrap

Severity: Informational

Description

Initial governance configuration temporarily relies on deployer-controlled setup before governance ownership is transferred to the Timelock.

Impact

Temporary centralization exists during bootstrap.

Recommendation

Transfer ownership immediately after deployment.

Status

Acknowledged.

S-05 — Governance Whale Voting Power

Severity: Informational

Description

Large token holders may exert disproportionate influence over governance proposals.

Impact

Governance outcomes may become centralized.

Recommendation

Future versions may implement delegation incentives or vote caps.

Status

Acknowledged.

S-06 — Repeated Storage Reads

Severity: Gas

Description

Some functions repeatedly access storage variables instead of caching them locally.

Impact

Minor gas inefficiencies.

Recommendation

Cache frequently used storage variables in memory.

Status

Acknowledged.

Centralization Analysis

The protocol uses OpenZeppelin TimelockController with a 2-day execution delay.

Governance execution requires:

- Proposal creation
- Voting period
- Queueing
- Timelock delay
- Execution

Role-based permissions are used for:

- Minting
- Oracle management
- Treasury operations
- Governance administration

Governance Attack Analysis

The protocol uses ERC20Votes snapshots, reducing same-block flash-loan governance manipulation risk.

Potential risks include:

- Whale domination
- Proposal spam
- Malicious governance upgrades

Mitigations include:

- Proposal threshold
- Quorum requirements
- Timelock delay

Oracle Attack Analysis

The protocol relies on Chainlink decentralized price feeds.

OracleAdapter reverts when stale data is detected.

Future improvements may include:

- Multiple oracle aggregation
- Circuit breakers
- Emergency pause mechanisms

Testing Summary

The project contains more than 80 automated tests.

Fuzz Testing

Fuzz testing validates:

- Vault accounting
- Governance voting
- Token transfers
- Oracle interactions

Invariant Testing

Invariant tests validate:

- Total supply conservation
- Treasury accounting
- Vault consistency

Fork Testing

Fork tests interact with:

- USDC
- Uniswap V2
- Chainlink price feeds

Static Analysis Results

Slither analysis reported:

- Zero High severity findings
- Zero Medium severity findings

Low and informational findings were reviewed and documented.

Conclusion

The RWA Tokenization Platform demonstrates strong focus on:

- Governance security
- Access control
- Oracle safety
- Treasury protection
- Automated testing
- Defensive engineering practices

The combination of:

- Unit testing
- Fuzz testing
- Invariant testing
- Fork testing
- Slither analysis

significantly improves protocol reliability and reduces attack surface.

Future improvements may include:

- Additional decentralization
- Advanced monitoring
- Upgradeable proxy architecture
- Expanded oracle redundancy
- Emergency pause mechanisms

Appendix

Tools Used

- Foundry
- Slither
- OpenZeppelin Contracts
- Chainlink

Coverage Summary

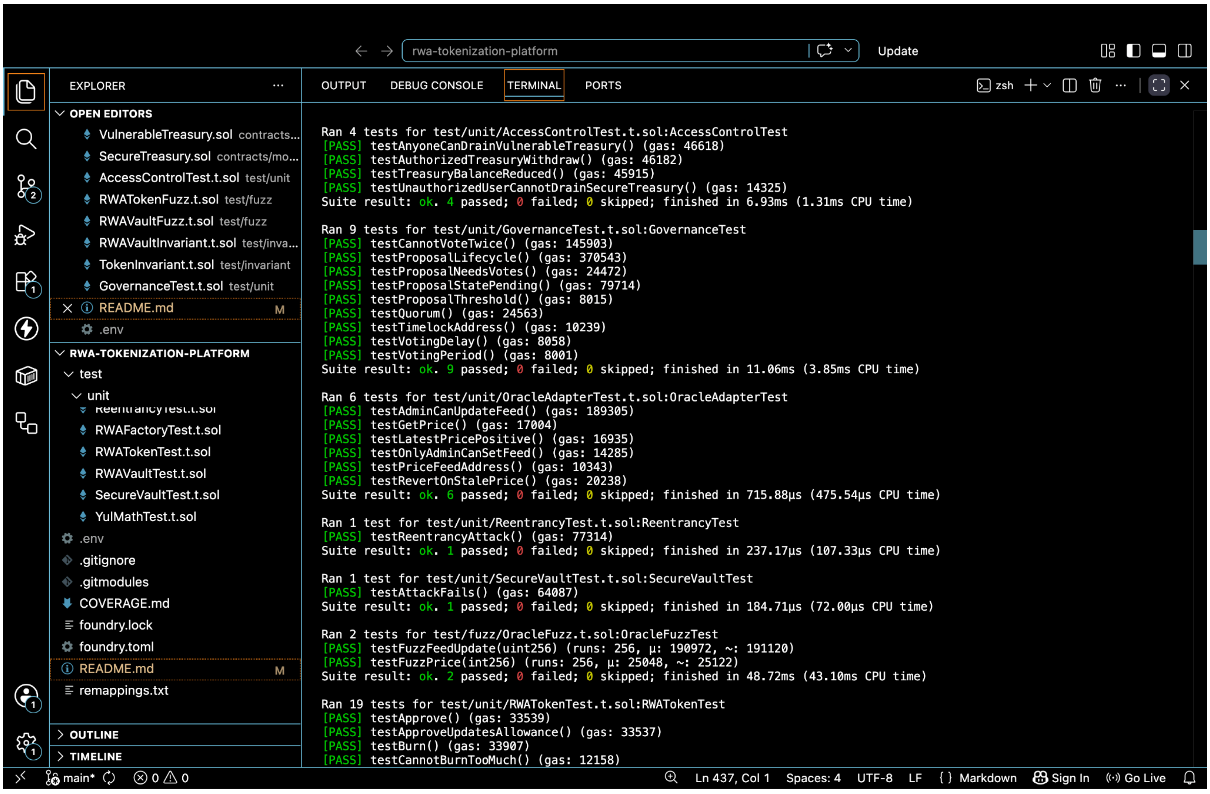
Metric	Result

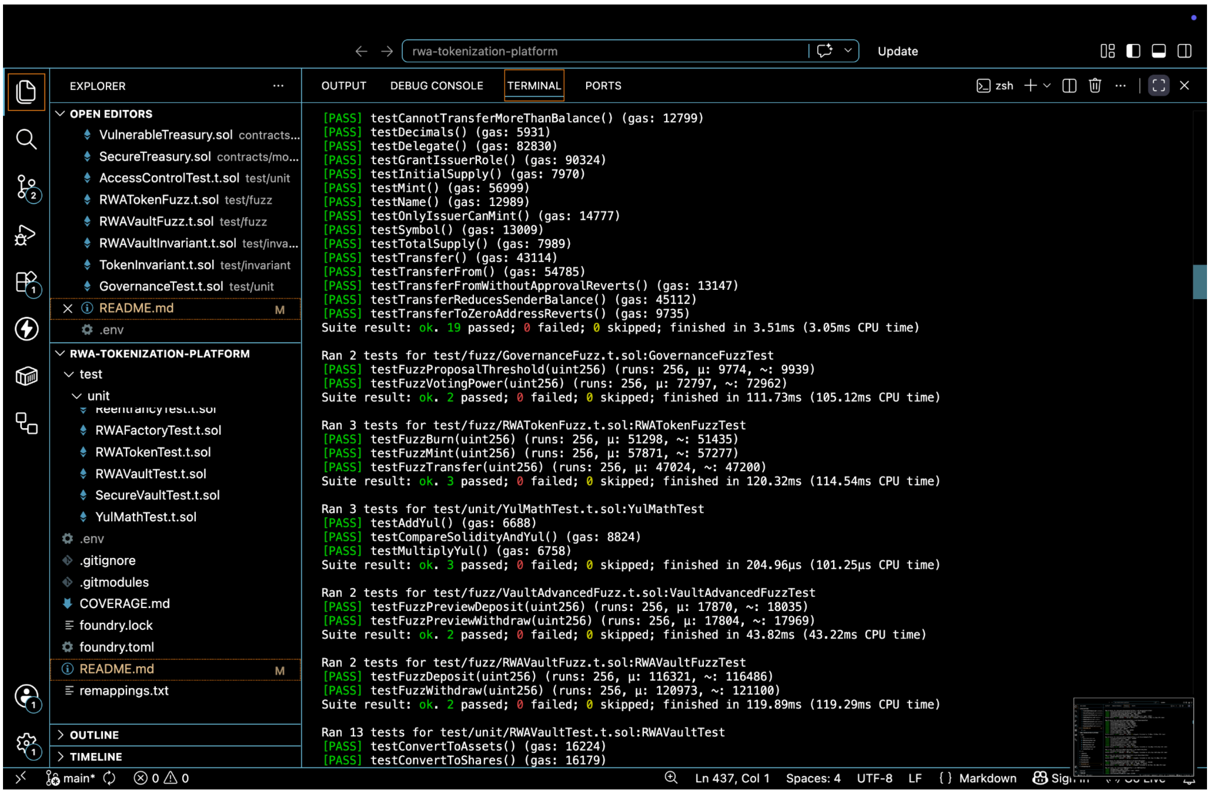
Line Coverage	94.21%
Statement Coverage	93.33%
Function Coverage	92.31%

CI Pipeline

GitHub Actions automatically runs:

- forge build
- forge test
- forge coverage
- Slither




rwa-tokenization-platform
Update
EXPLORER
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
zsh
[PASS] testCannotTransferMoreThanBalance() (gas: 12799)
[PASS] testDecimals() (gas: 5931)
[PASS] testDelegate() (gas: 82830)
[PASS] testGrantIssuerRole() (gas: 90324)
[PASS] testInitialSupply() (gas: 7970)
[PASS] testMint() (gas: 56999)
[PASS] testName() (gas: 12989)
[PASS] testOnlyIssuerCanMint() (gas: 14777)
[PASS] testSymbol() (gas: 13009)
[PASS] testTotalSupply() (gas: 7989)
[PASS] testTransfer() (gas: 43114)
[PASS] testTransferFrom() (gas: 54785)
[PASS] testTransferFromWithoutApprovalReverts() (gas: 13147)
[PASS] testTransferReducesSenderBalance() (gas: 45112)
[PASS] testTransferToZeroAddressReverts() (gas: 9735)
Suite result: ok. 19 passed; 0 failed; 0 skipped; finished in 3.51ms (3.05ms CPU time)
Ran 2 tests for test/fuzz/GovernanceFuzz.t.sol:GovernanceFuzzTest
[PASS] testFuzzProposalThreshold(uint256) (runs: 256, μ: 9774, ~: 9939)
[PASS] testFuzzVotingPower(uint256) (runs: 256, μ: 72797, ~: 72962)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 111.73ms (105.12ms CPU time)
Ran 3 tests for test/fuzz/RWATokenFuzz.t.sol:RWATokenFuzzTest
[PASS] testFuzzBurn(uint256) (runs: 256, μ: 51298, ~: 51435)
[PASS] testFuzzMint(uint256) (runs: 256, μ: 57871, ~: 57277)
[PASS] testFuzzTransfer(uint256) (runs: 256, μ: 47024, ~: 47200)
Suite result: ok. 3 passed; 0 failed; 0 skipped; finished in 120.32ms (114.54ms CPU time)
Ran 3 tests for test/unit/YulMathTest.t.sol:YulMathTest
[PASS] testAddYul() (gas: 6688)
[PASS] testCompareSolidityAndYul() (gas: 8824)
[PASS] testMultiplyYul() (gas: 6758)
Suite result: ok. 3 passed; 0 failed; 0 skipped; finished in 204.96μs (101.25μs CPU time)
Ran 2 tests for test/fuzz/VaultAdvancedFuzz.t.sol:VaultAdvancedFuzzTest
[PASS] testFuzzPreviewDeposit(uint256) (runs: 256, μ: 17870, ~: 18035)
[PASS] testFuzzPreviewWithdraw(uint256) (runs: 256, μ: 17804, ~: 17969)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 43.82ms (43.22ms CPU time)
Ran 2 tests for test/fuzz/RWAVaultFuzz.t.sol:RWAVaultFuzzTest
[PASS] testFuzzDeposit(uint256) (runs: 256, μ: 116321, ~: 116486)
[PASS] testFuzzWithdraw(uint256) (runs: 256, μ: 120973, ~: 121100)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 119.89ms (119.29ms CPU time)
Ran 13 tests for test/unit/RWAVaultTest.t.sol:RWAVaultTest
[PASS] testConvertToAssets() (gas: 16224)
[PASS] testConvertToShares() (gas: 16179)
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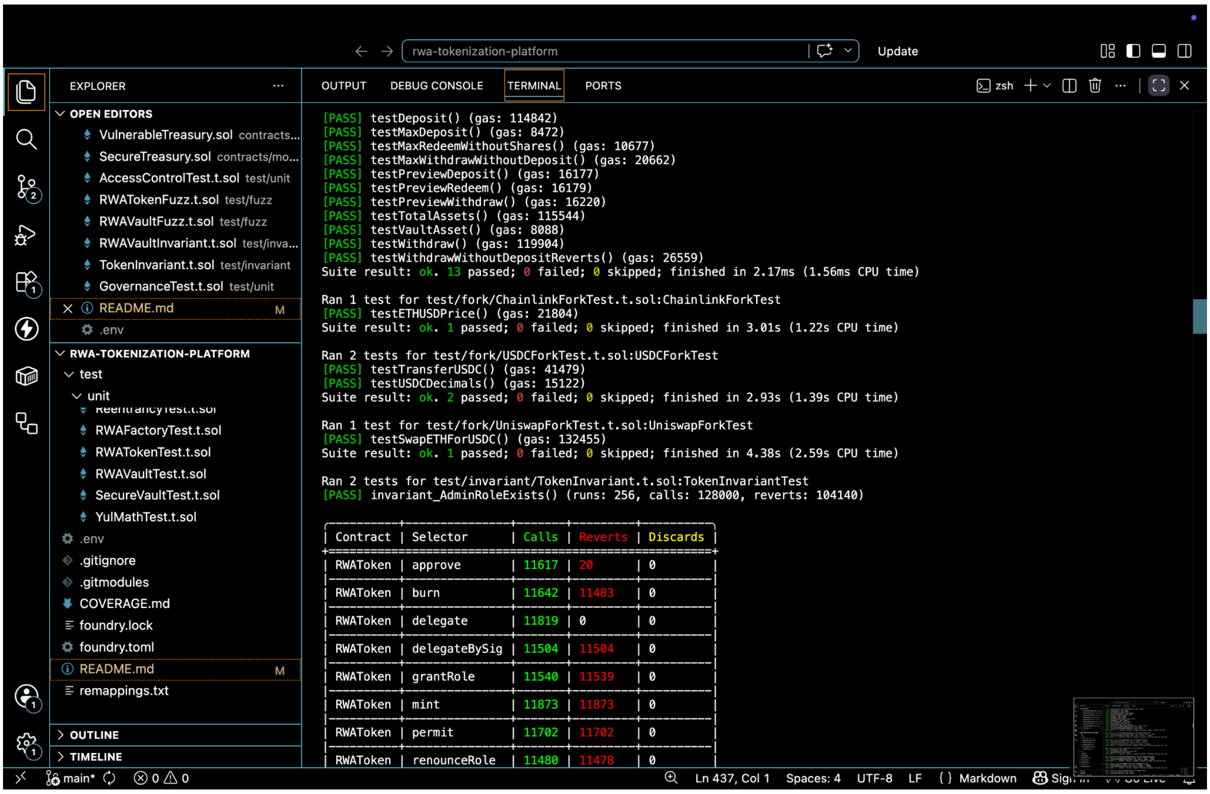
rwa-tokenization-platform
Update
EXPLORER
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
zsh
[PASS] testDeposit() (gas: 114842)
[PASS] testMaxDeposit() (gas: 8472)
[PASS] testMaxRedeemWithoutShares() (gas: 10677)
[PASS] testMaxWithdrawWithoutDeposit() (gas: 20662)
[PASS] testPreviewDeposit() (gas: 16177)
[PASS] testPreviewRedeem() (gas: 16179)
[PASS] testPreviewWithdraw() (gas: 16220)
[PASS] testTotalAssets() (gas: 115544)
[PASS] testVaultAsset() (gas: 8088)
[PASS] testWithdraw() (gas: 119904)
[PASS] testWithdrawWithoutDepositReverts() (gas: 26559)
Suite result: ok. 13 passed; 0 failed; 0 skipped; finished in 2.17ms (1.56ms CPU time)
Ran 1 test for test/fork/ChainlinkForkTest.t.sol:ChainlinkForkTest
[PASS] testETHUSDPrice() (gas: 21804)
Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 3.01s (1.22s CPU time)
Ran 2 tests for test/fork/USDCForkTest.t.sol:USDCForkTest
[PASS] testTransferUSDC() (gas: 41479)
[PASS] testUSDCDecimals() (gas: 15122)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 2.93s (1.39s CPU time)
Ran 1 test for test/fork/UniswapForkTest.t.sol:UniswapForkTest
[PASS] testSwapETHForUSDC() (gas: 132455)
Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 4.38s (2.59s CPU time)
Ran 2 tests for test/invariant/TokenInvariant.t.sol:TokenInvariantTest
[PASS] invariant_AdminRoleExists() (runs: 256, calls: 128000, reverts: 104140)
Contract	Selector	Calls	Reverts	Discards
RWAToken	approve	11617	20	0
RWAToken	burn	11642	11483	0
RWAToken	delegate	11819	0	0
RWAToken	delegateBySig	11504	11504	0
RWAToken	grantRole	11540	11539	0
RWAToken	mint	11873	11873	0
RWAToken	permit	11702	11702	0
RWAToken	renounceRole	11480	11478	0
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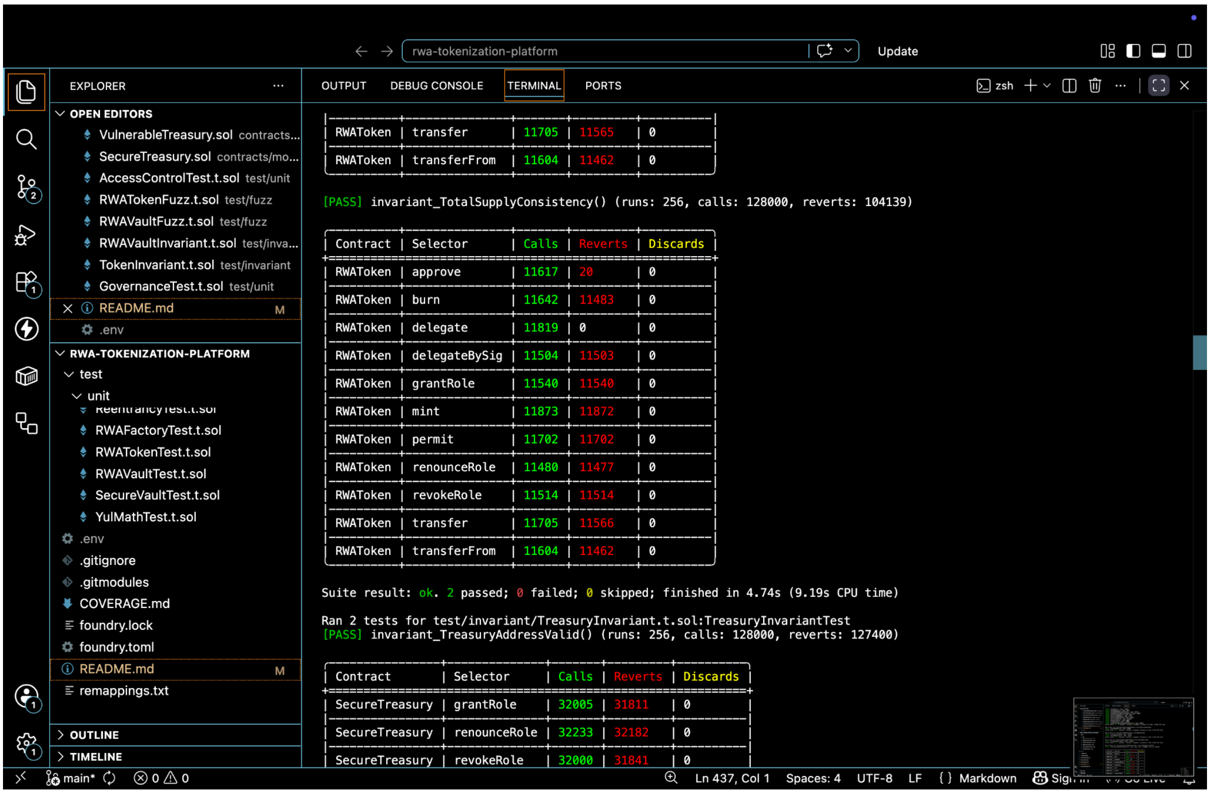
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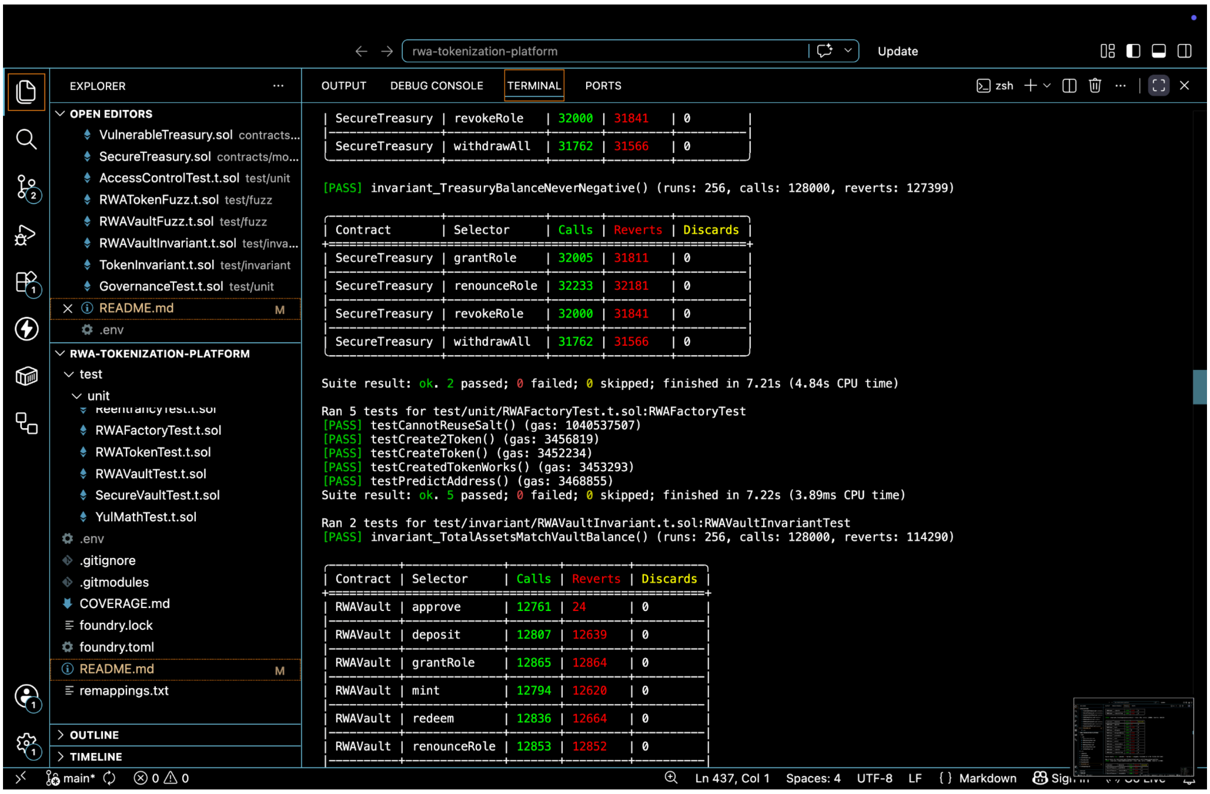
EXPLORER
OPEN EDITORS
OUTPUT DEBUG CONSOLE TERMINAL PORTS
| RWAToken | transfer | 11705 | 11565 | 0 |
| RWAToken | transferFrom | 11604 | 11462 | 0 |
[PASS] invariant_TotalSupplyConsistency() (runs: 256, calls: 128000, reverts: 104139)
Contract	Selector	Calls	Reverts	Discards
RWAToken	approve	11617	20	0
RWAToken	burn	11642	11483	0
RWAToken	delegate	11819	0	0
RWAToken	delegateBySig	11504	11503	0
RWAToken	grantRole	11540	11540	0
RWAToken	mint	11873	11872	0
RWAToken	permit	11702	11702	0
RWAToken	renounceRole	11480	11477	0
RWAToken	revokeRole	11514	11514	0
RWAToken	transfer	11705	11566	0
RWAToken	transferFrom	11604	11462	0
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 4.74s (9.19s CPU time)
Ran 2 tests for test/invariant/TreasuryInvariant.t.sol:TreasuryInvariantTest
[PASS] invariant_TreasuryAddressValid() (runs: 256, calls: 128000, reverts: 127400)
Contract	Selector	Calls	Reverts	Discards
SecureTreasury	grantRole	32005	31811	0
SecureTreasury	renounceRole	32233	32182	0
SecureTreasury	revokeRole	32000	31841	0
main* Ln 437, Col 1 Spaces: 4 UTF-8 LF Markdown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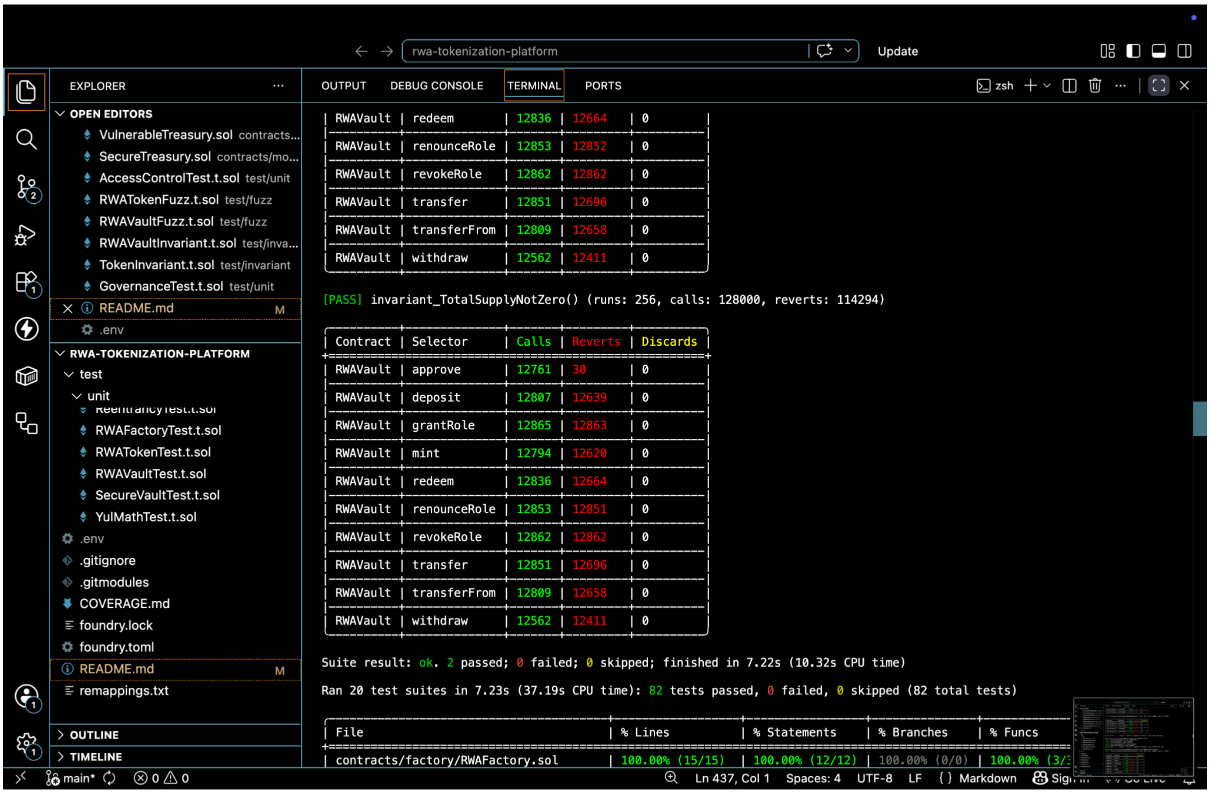
EXPLORER
OPEN EDITORS
OUTPUT DEBUG CONSOLE TERMINAL PORTS
| SecureTreasury | revokeRole | 32000 | 31841 | 0 |
| SecureTreasury | withdrawAll | 31762 | 31566 | 0 |
[PASS] invariant_TreasuryBalanceNeverNegative() (runs: 256, calls: 128000, reverts: 127399)
Contract	Selector	Calls	Reverts	Discards
SecureTreasury	grantRole	32005	31811	0
SecureTreasury	renounceRole	32233	32181	0
SecureTreasury	revokeRole	32000	31841	0
SecureTreasury	withdrawAll	31762	31566	0
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 7.21s (4.84s CPU time)
Ran 5 tests for test/unit/RWAFactoryTest.t.sol:RWAFactoryTest
[PASS] testCannotReuseSalt() (gas: 1040537507)
[PASS] testCreate2Token() (gas: 3456819)
[PASS] testCreateToken() (gas: 3452234)
[PASS] testCreatedTokenWorks() (gas: 3453293)
[PASS] testPredictAddress() (gas: 3468855)
Suite result: ok. 5 passed; 0 failed; 0 skipped; finished in 7.22s (3.89ms CPU time)
Ran 2 tests for test/invariant/RWAVaultInvariant.t.sol:RWAVaultInvariantTest
[PASS] invariant_TotalAssetsMatchVaultBalance() (runs: 256, calls: 128000, reverts: 114290)
Contract	Selector	Calls	Reverts	Discards
RWAVault	approve	12761	24	0
RWAVault	deposit	12807	12639	0
RWAVault	grantRole	12865	12864	0
RWAVault	mint	12794	12620	0
RWAVault	redeem	12836	12664	0
RWAVault	renounceRole	12853	12852	0
main* Ln 437, Col 1 Spaces: 4 UTF-8 LF Markdown


rwa-tokenization-platform
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VulnerableTreasury.sol
SecureTreasury.sol
AccessControlTest.t.sol test/unit
RWATokenFuzz.t.sol test/fuzz
RWAVaultFuzz.t.sol test/fuzz
RWAVaultInvariant.t.sol
TokenInvariant.t.sol test/invariant
GovernanceTest.t.sol test/unit
README.md
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RWA-TOKENIZATION-PLATFORM
test
unit
RWAFactoryTest.t.sol
RWATokenTest.t.sol
RWAVaultTest.t.sol
SecureVaultTest.t.sol
YulMathTest.t.sol
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COVERAGE.md
foundry.lock
foundry.toml
README.md
remappings.txt
OUTLINE
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RWAVault	redeem	12836	12664	0
RWAVault	renounceRole	12853	12852	0
RWAVault	revokeRole	12862	12862	0
RWAVault	transfer	12851	12696	0
RWAVault	transferFrom	12809	12658	0
RWAVault	withdraw	12562	12411	0
[PASS] invariant_TotalSupplyNotZero() (runs: 256, calls: 128000, reverts: 114294)
Contract	Selector	Calls	Reverts	Discards
RWAVault	approve	12761	30	0
RWAVault	deposit	12807	12639	0
RWAVault	grantRole	12865	12863	0
RWAVault	mint	12794	12620	0
RWAVault	redeem	12836	12664	0
RWAVault	renounceRole	12853	12851	0
RWAVault	revokeRole	12862	12862	0
RWAVault	transfer	12851	12696	0
RWAVault	transferFrom	12809	12658	0
RWAVault	withdraw	12562	12411	0
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 7.22s (10.32s CPU time)
Ran 20 test suites in 7.23s (37.19s CPU time): 82 tests passed, 0 failed, 0 skipped (82 total tests)
File	% Lines	% Statements	% Branches	% Funcs
contracts/factory/RWAFactory.sol	100.00% (15/15)	100.00% (12/12)	100.00% (0/0)	100.00% (3/
main* 0 0
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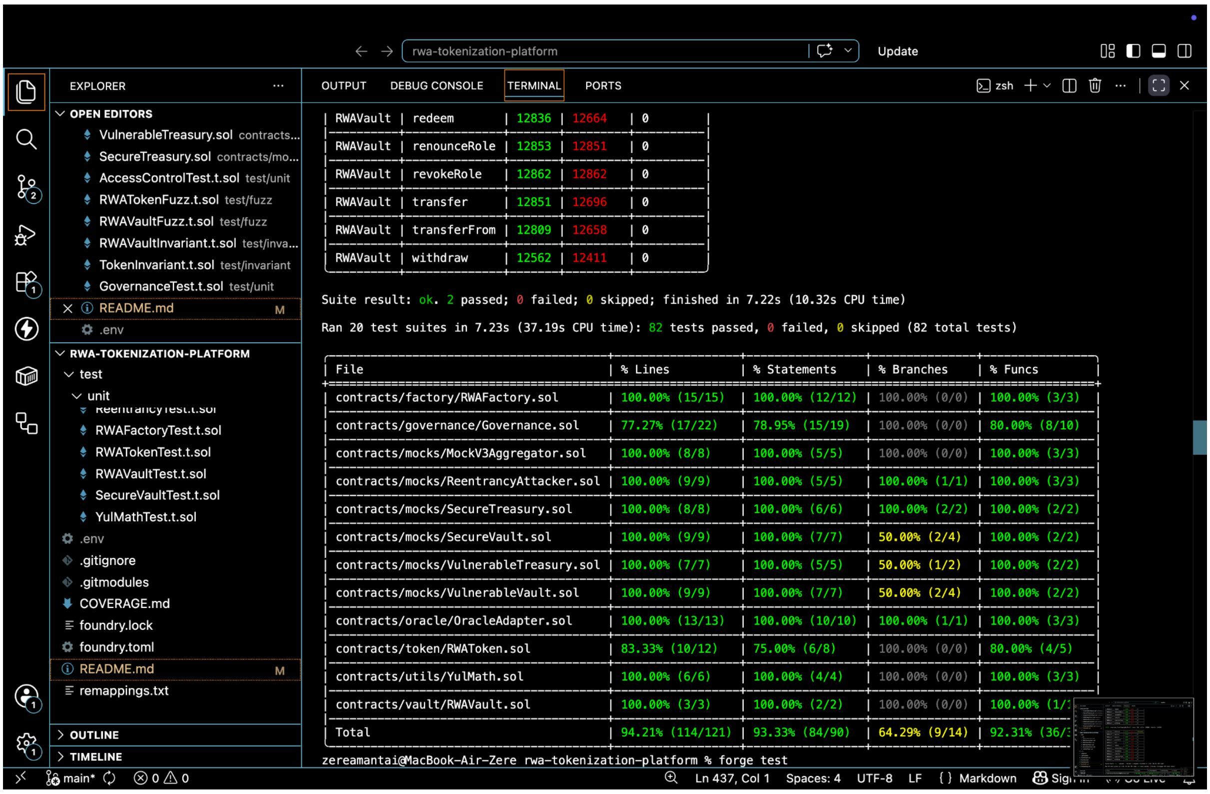
rwa-tokenization-platform
EXPLORER
OPEN EDITORS
VulnerableTreasury.sol
SecureTreasury.sol
AccessControlTest.t.sol test/unit
RWATokenFuzz.t.sol test/fuzz
RWAVaultFuzz.t.sol test/fuzz
RWAVaultInvariant.t.sol
TokenInvariant.t.sol test/invariant
GovernanceTest.t.sol test/unit
README.md
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RWA-TOKENIZATION-PLATFORM
test
unit
RWAFactoryTest.t.sol
RWATokenTest.t.sol
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RWAVault	redeem	12836	12664	0
RWAVault	renounceRole	12853	12851	0
RWAVault	revokeRole	12862	12862	0
RWAVault	transfer	12851	12696	0
RWAVault	transferFrom	12809	12658	0
RWAVault	withdraw	12562	12411	0
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 7.22s (10.32s CPU time)
Ran 20 test suites in 7.23s (37.19s CPU time): 82 tests passed, 0 failed, 0 skipped (82 total tests)
File	% Lines	% Statements	% Branches	% Funcs
contracts/factory/RWAFactory.sol	100.00% (15/15)	100.00% (12/12)	100.00% (0/0)	100.00% (3/3)
contracts/governance/Governance.sol	77.27% (17/22)	78.95% (15/19)	100.00% (0/0)	80.00% (8/10)
contracts/mocks/MockV3Aggregator.sol	100.00% (8/8)	100.00% (5/5)	100.00% (0/0)	100.00% (3/3)
contracts/mocks/ReentrancyAttacker.sol	100.00% (9/9)	100.00% (5/5)	100.00% (1/1)	100.00% (3/3)
contracts/mocks/SecureTreasury.sol	100.00% (8/8)	100.00% (6/6)	100.00% (2/2)	100.00% (2/2)
contracts/mocks/SecureVault.sol	100.00% (9/9)	100.00% (7/7)	50.00% (2/4)	100.00% (2/2)
contracts/mocks/VulnerableTreasury.sol	100.00% (7/7)	100.00% (5/5)	50.00% (1/2)	100.00% (2/2)
contracts/mocks/VulnerableVault.sol	100.00% (9/9)	100.00% (7/7)	50.00% (2/4)	100.00% (2/2)
contracts/oracle/OracleAdapter.sol	100.00% (13/13)	100.00% (10/10)	100.00% (1/1)	100.00% (3/3)
contracts/token/RWAToken.sol	83.33% (10/12)	75.00% (6/8)	100.00% (0/0)	80.00% (4/5)
contracts/utils/YulMath.sol	100.00% (6/6)	100.00% (4/4)	100.00% (0/0)	100.00% (3/3)
contracts/vault/RWAVault.sol	100.00% (3/3)	100.00% (2/2)	100.00% (0/0)	100.00% (1/
Total	94.21% (114/121)	93.33% (84/90)	64.29% (9/14)	92.31% (36/
zereamantai@MacBook-Air-Zere rwa-tokenization-platform % forge test
main* 0 0
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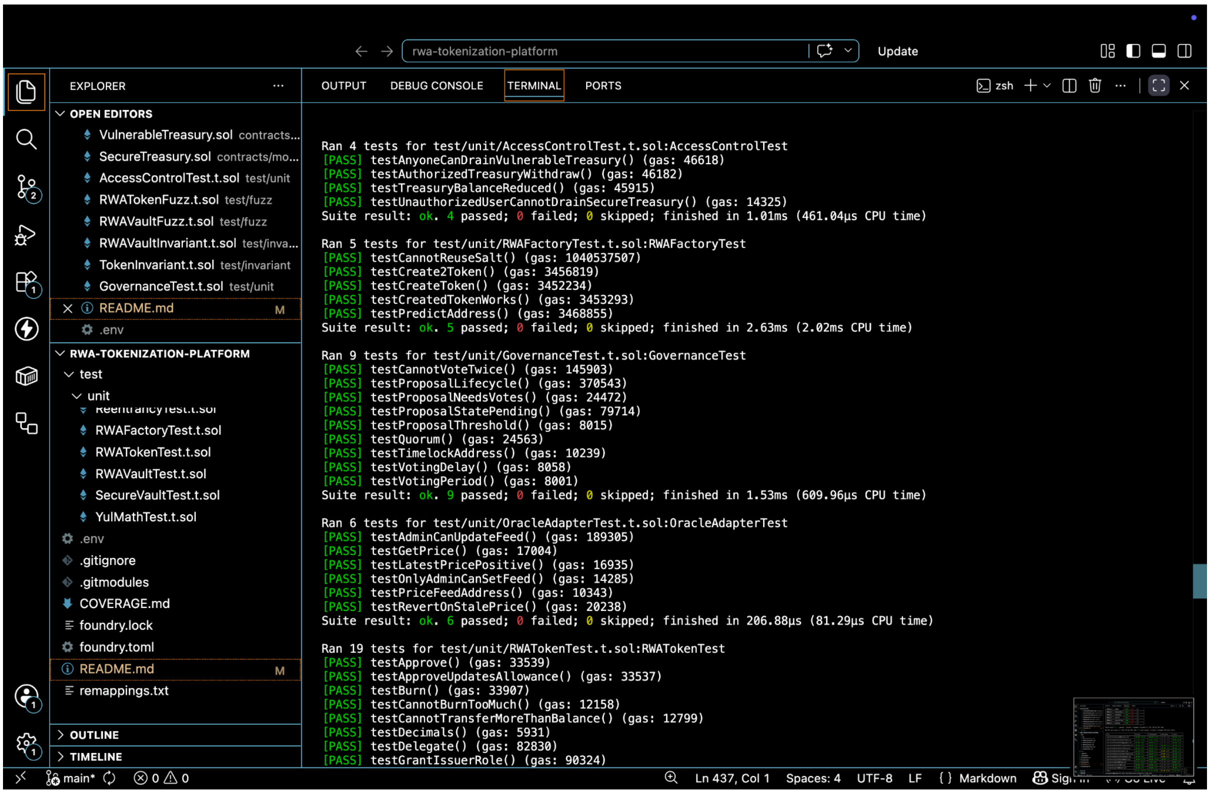
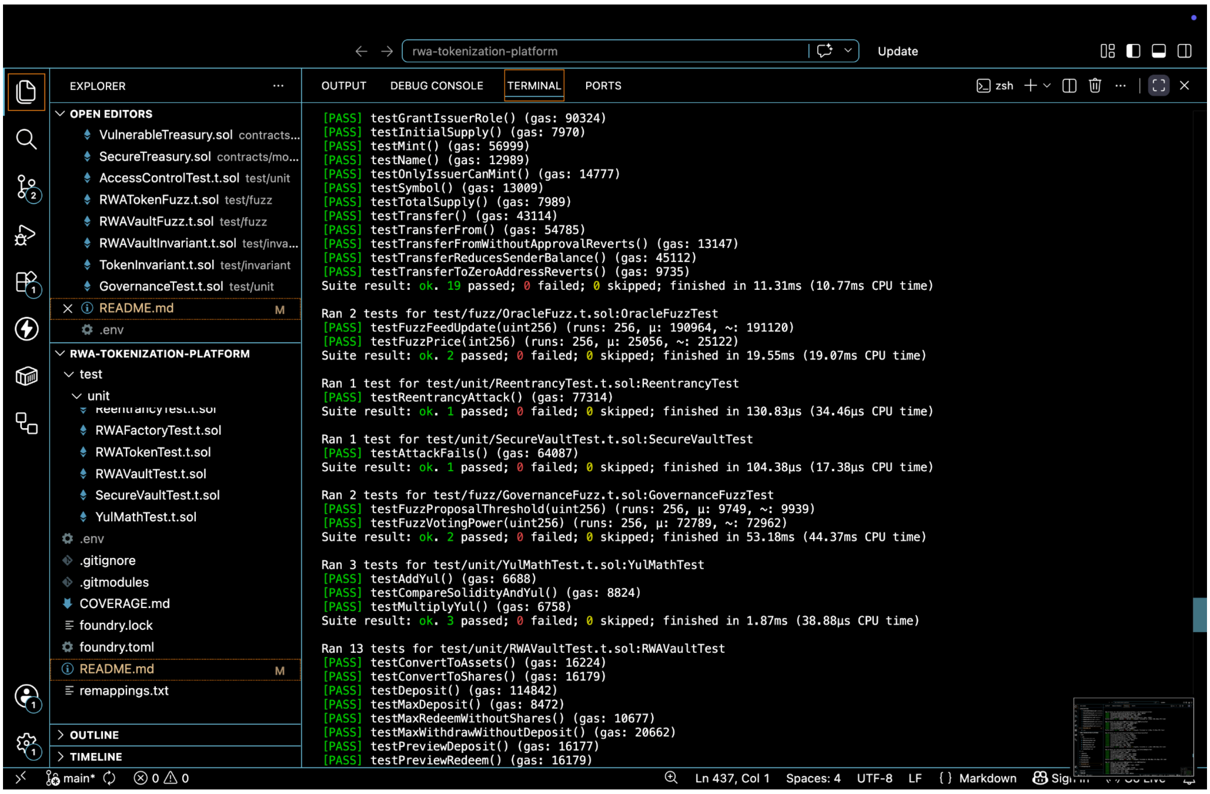
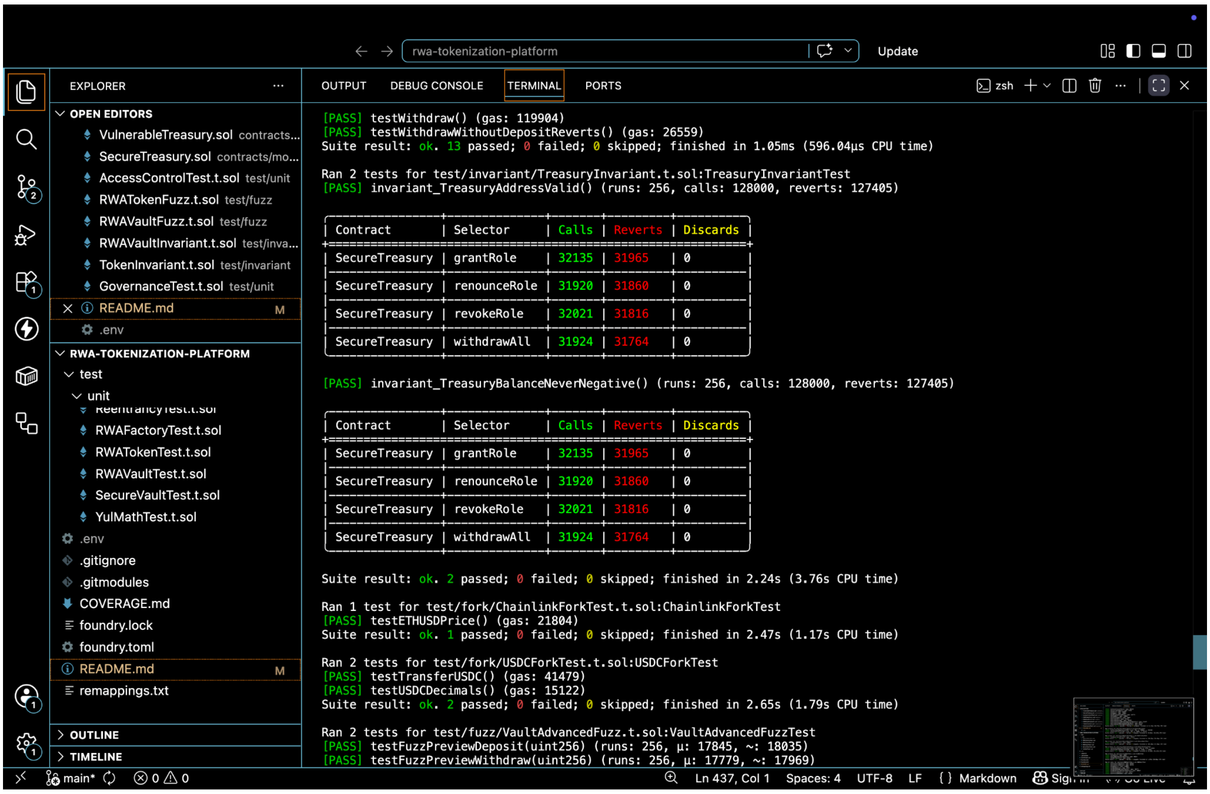

rwa-tokenization-platform
EXPLORER
OUTPUT DEBUG CONSOLE TERMINAL PORTS
Ran 4 tests for test/unit/AccessControlTest.t.sol:AccessControlTest
[PASS] testAnyoneCanDrainVulnerableTreasury() (gas: 46618)
[PASS] testAuthorizedTreasuryWithdraw() (gas: 46182)
[PASS] testTreasuryBalanceReduced() (gas: 45915)
[PASS] testUnauthorizedUserCannotDrainSecureTreasury() (gas: 14325)
Suite result: ok. 4 passed; 0 failed; 0 skipped; finished in 1.01ms (461.04µs CPU time)
Ran 5 tests for test/unit/RWAFactoryTest.t.sol:RWAFactoryTest
[PASS] testCannotReuseSalt() (gas: 1040537507)
[PASS] testCreate2Token() (gas: 3456819)
[PASS] testCreateToken() (gas: 3452234)
[PASS] testCreatedTokenWorks() (gas: 3453293)
[PASS] testPredictAddress() (gas: 3468855)
Suite result: ok. 5 passed; 0 failed; 0 skipped; finished in 2.63ms (2.02ms CPU time)
Ran 9 tests for test/unit/GovernanceTest.t.sol:GovernanceTest
[PASS] testCannotVoteTwice() (gas: 145903)
[PASS] testProposalLifecycle() (gas: 370543)
[PASS] testProposalNeedsVotes() (gas: 24472)
[PASS] testProposalStatePending() (gas: 79714)
[PASS] testProposalThreshold() (gas: 8015)
[PASS] testQuorum() (gas: 24563)
[PASS] testTimelockAddress() (gas: 10239)
[PASS] testVotingDelay() (gas: 8058)
[PASS] testVotingPeriod() (gas: 8001)
Suite result: ok. 9 passed; 0 failed; 0 skipped; finished in 1.53ms (609.96µs CPU time)
Ran 6 tests for test/unit/OracleAdapterTest.t.sol:OracleAdapterTest
[PASS] testAdminCanUpdateFeed() (gas: 189305)
[PASS] testGetPrice() (gas: 17004)
[PASS] testLatestPricePositive() (gas: 16935)
[PASS] testOnlyAdminCanSetFeed() (gas: 14285)
[PASS] testPriceFeedAddress() (gas: 10343)
[PASS] testRevertOnStalePrice() (gas: 20238)
Suite result: ok. 6 passed; 0 failed; 0 skipped; finished in 206.88µs (81.29µs CPU time)
Ran 19 tests for test/unit/RWATokenTest.t.sol:RWATokenTest
[PASS] testApprove() (gas: 33539)
[PASS] testApproveUpdatesAllowance() (gas: 33537)
[PASS] testBurn() (gas: 33907)
[PASS] testCannotBurnTooMuch() (gas: 12158)
[PASS] testCannotTransferMoreThanBalance() (gas: 12799)
[PASS] testDecimals() (gas: 5931)
[PASS] testDelegate() (gas: 82830)
[PASS] testGrantIssuerRole() (gas: 90324)
main* Ln 437, Col 1 Spaces: 4 UTF-8 LF Markdown


rwa-tokenization-platform
EXPLORER
OUTPUT DEBUG CONSOLE TERMINAL PORTS
[PASS] testGrantIssuerRole() (gas: 90324)
[PASS] testInitialSupply() (gas: 7970)
[PASS] testMint() (gas: 56999)
[PASS] testName() (gas: 12989)
[PASS] testOnlyIssuerCanMint() (gas: 14777)
[PASS] testSymbol() (gas: 13009)
[PASS] testTotalSupply() (gas: 7989)
[PASS] testTransfer() (gas: 43114)
[PASS] testTransferFrom() (gas: 54785)
[PASS] testTransferFromWithoutApprovalReverts() (gas: 13147)
[PASS] testTransferReducesSenderBalance() (gas: 45112)
[PASS] testTransferToZeroAddressReverts() (gas: 9735)
Suite result: ok. 19 passed; 0 failed; 0 skipped; finished in 11.31ms (10.77ms CPU time)
Ran 2 tests for test/fuzz/OracleFuzz.t.sol:OracleFuzzTest
[PASS] testFuzzFeedUpdate(int256) (runs: 256, µ: 190964, ~: 191120)
[PASS] testFuzzPrice(int256) (runs: 256, µ: 25056, ~: 25122)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 19.55ms (19.07ms CPU time)
Ran 1 test for test/unit/ReentrancyTest.t.sol:ReentrancyTest
[PASS] testReentrancyAttack() (gas: 77314)
Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 130.83µs (34.46µs CPU time)
Ran 1 test for test/unit/SecureVaultTest.t.sol:SecureVaultTest
[PASS] testAttackFails() (gas: 64087)
Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 104.38µs (17.38µs CPU time)
Ran 2 tests for test/fuzz/GovernanceFuzz.t.sol:GovernanceFuzzTest
[PASS] testFuzzProposalThreshold(uint256) (runs: 256, µ: 9749, ~: 9939)
[PASS] testFuzzVotingPower(uint256) (runs: 256, µ: 72789, ~: 72962)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 53.18ms (44.37ms CPU time)
Ran 3 tests for test/unit/YulMathTest.t.sol:YulMathTest
[PASS] testAddYul() (gas: 6688)
[PASS] testCompareSolidityAndYul() (gas: 8824)
[PASS] testMultiplyYul() (gas: 6758)
Suite result: ok. 3 passed; 0 failed; 0 skipped; finished in 1.87ms (38.88µs CPU time)
Ran 13 tests for test/unit/RWAVaultTest.t.sol:RWAVaultTest
[PASS] testConvertToAssets() (gas: 16224)
[PASS] testConvertToShares() (gas: 16179)
[PASS] testDeposit() (gas: 114842)
[PASS] testMaxDeposit() (gas: 8472)
[PASS] testMaxRedeemWithoutShares() (gas: 10677)
[PASS] testMaxWithdrawWithoutDeposit() (gas: 20662)
[PASS] testPreviewDeposit() (gas: 16177)
[PASS] testPreviewRedeem() (gas: 16179)
main* Ln 437, Col 1 Spaces: 4 UTF-8 LF Markdown


rwa-tokenization-platform
Update
EXPLORER
OPEN EDITORS
VulnerableTreasury.sol contracts...
SecureTreasury.sol contracts/mo...
AccessControlTest.t.sol test/unit
RWATokenFuzz.t.sol test/fuzz
RWAVaultFuzz.t.sol test/fuzz
RWAVaultInvariant.t.sol test/inva...
TokenInvariant.t.sol test/invariant
GovernanceTest.t.sol test/unit
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RWA-TOKENIZATION-PLATFORM
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unit
RWAFactoryTest.t.sol
RWATokenTest.t.sol
RWAVaultTest.t.sol
SecureVaultTest.t.sol
YulMathTest.t.sol
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remappings.txt
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[PASS] testWithdraw() (gas: 119904)
[PASS] testWithdrawWithoutDepositReverts() (gas: 26559)
Suite result: ok. 13 passed; 0 failed; 0 skipped; finished in 1.05ms (596.04µs CPU time)
Ran 2 tests for test/invariant/TreasuryInvariant.t.sol:TreasuryInvariantTest
[PASS] invariant_TreasuryAddressValid() (runs: 256, calls: 128000, reverts: 127405)
Contract	Selector	Calls	Reverts	Discards
SecureTreasury	grantRole	32135	31965	0
SecureTreasury	renounceRole	31920	31860	0
SecureTreasury	revokeRole	32021	31816	0
SecureTreasury	withdrawAll	31924	31764	0
[PASS] invariant_TreasuryBalanceNeverNegative() (runs: 256, calls: 128000, reverts: 127405)
Contract	Selector	Calls	Reverts	Discards
SecureTreasury	grantRole	32135	31965	0
SecureTreasury	renounceRole	31920	31860	0
SecureTreasury	revokeRole	32021	31816	0
SecureTreasury	withdrawAll	31924	31764	0
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 2.24s (3.76s CPU time)
Ran 1 test for test/fork/ChainlinkForkTest.t.sol:ChainlinkForkTest
[PASS] testETHUSDPrice() (gas: 21804)
Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 2.47s (1.17s CPU time)
Ran 2 tests for test/fork/USDCForkTest.t.sol:USDCForkTest
[PASS] testTransferUSDC() (gas: 41479)
[PASS] testUSDCDecimals() (gas: 15122)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 2.65s (1.79s CPU time)
Ran 2 tests for test/fuzz/VaultAdvancedFuzz.t.sol:VaultAdvancedFuzzTest
[PASS] testFuzzPreviewDeposit(uint256) (runs: 256, µ: 17845, ~: 18035)
[PASS] testFuzzPreviewWithdraw(uint256) (runs: 256, µ: 17779, ~: 17969)
main*
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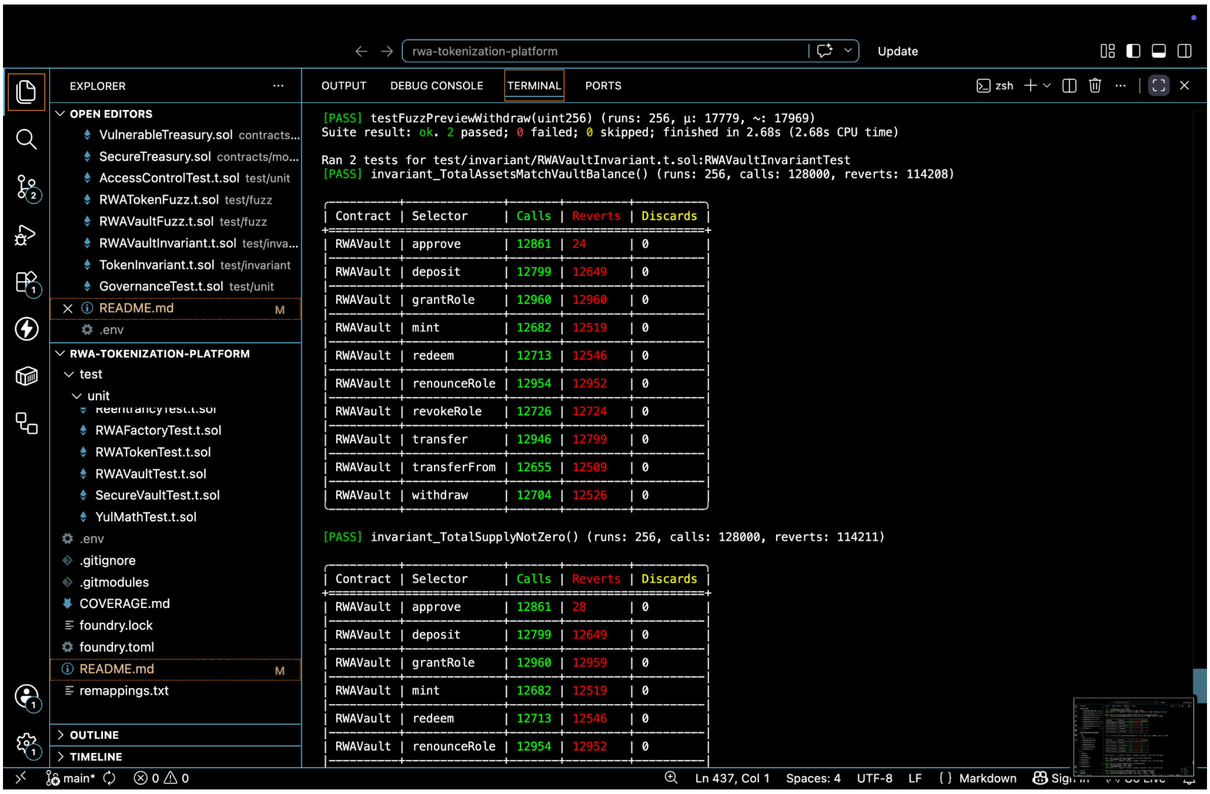
Ln 437, Col 1 Spaces: 4 UTF-8 LF Markdown


rwa-tokenization-platform
Update
TERMINAL
[PASS] testFuzzPreviewWithdraw(uint256) (runs: 256, µ: 17779, ~: 17969)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 2.68s (2.68s CPU time)
Ran 2 tests for test/invariant/RWAVaultInvariant.t.sol:RWAVaultInvariantTest
[PASS] invariant_TotalAssetsMatchVaultBalance() (runs: 256, calls: 128000, reverts: 114208)
Contract	Selector	Calls	Reverts	Discards
RWAVault	approve	12861	24	0
RWAVault	deposit	12799	12649	0
RWAVault	grantRole	12960	12960	0
RWAVault	mint	12682	12519	0
RWAVault	redeem	12713	12546	0
RWAVault	renounceRole	12954	12952	0
RWAVault	revokeRole	12726	12724	0
RWAVault	transfer	12946	12799	0
RWAVault	transferFrom	12655	12509	0
RWAVault	withdraw	12704	12526	0
[PASS] invariant_TotalSupplyNotZero() (runs: 256, calls: 128000, reverts: 114211)
Contract	Selector	Calls	Reverts	Discards
RWAVault	approve	12861	28	0
RWAVault	deposit	12799	12649	0
RWAVault	grantRole	12960	12959	0
RWAVault	mint	12682	12519	0
RWAVault	redeem	12713	12546	0
RWAVault	renounceRole	12954	12952	0
main*
0 0
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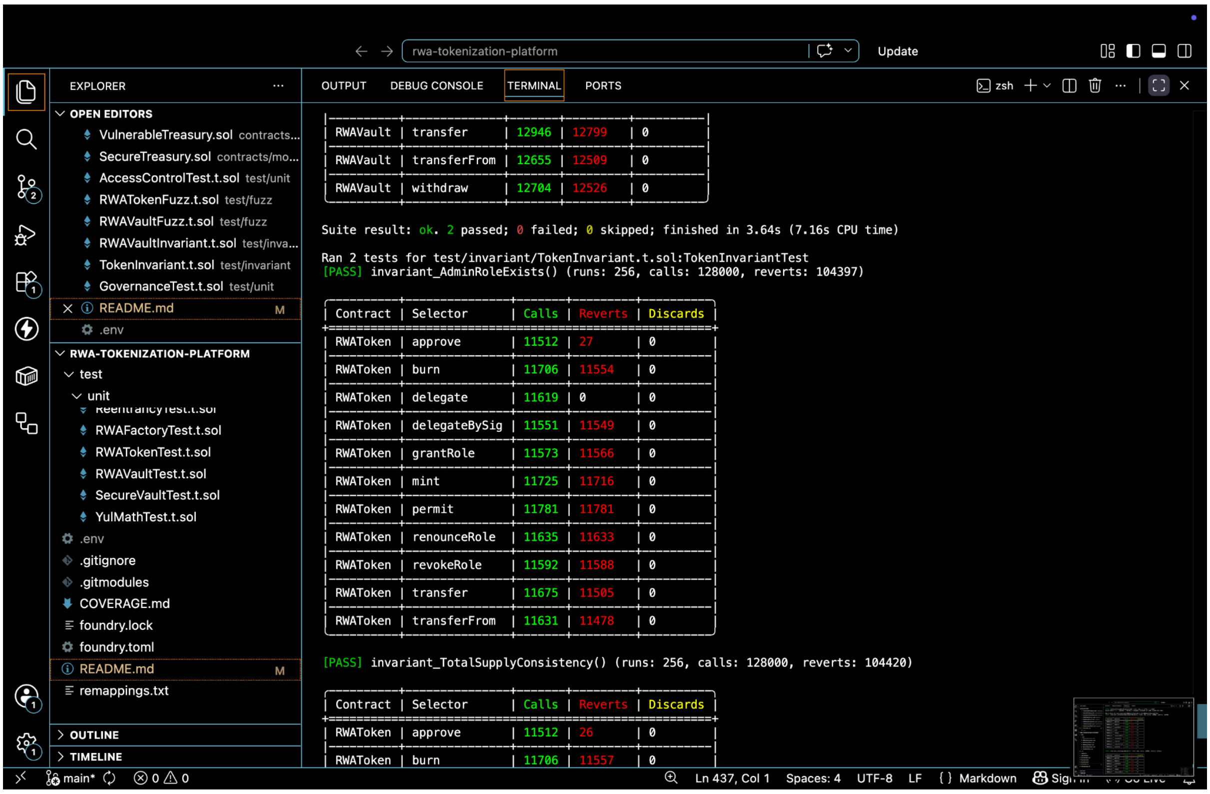
rwa-tokenization-platform
Update
EXPLORER
OPEN EDITORS
VulnerableTreasury.sol contracts...
SecureTreasury.sol contracts/mo...
AccessControlTest.t.sol test/unit
RWATokenFuzz.t.sol test/fuzz
RWAVaultFuzz.t.sol test/fuzz
RWAVaultInvariant.t.sol test/inva...
TokenInvariant.t.sol test/invariant
GovernanceTest.t.sol test/unit
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RWA-TOKENIZATION-PLATFORM
test
unit
RWAFactoryTest.t.sol
RWATokenTest.t.sol
RWAVaultTest.t.sol
SecureVaultTest.t.sol
YulMathTest.t.sol
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RWAVault	transfer	12946	12799	0
RWAVault	transferFrom	12655	12509	0
RWAVault	withdraw	12704	12526	0
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 3.64s (7.16s CPU time)
Ran 2 tests for test/invariant/TokenInvariant.t.sol:TokenInvariantTest
[PASS] invariant_AdminRoleExists() (runs: 256, calls: 128000, reverts: 104397)
Contract	Selector	Calls	Reverts	Discards
RWAToken	approve	11512	27	0
RWAToken	burn	11706	11554	0
RWAToken	delegate	11619	0	0
RWAToken	delegateBySig	11551	11549	0
RWAToken	grantRole	11573	11566	0
RWAToken	mint	11725	11716	0
RWAToken	permit	11781	11781	0
RWAToken	renounceRole	11635	11633	0
RWAToken	revokeRole	11592	11588	0
RWAToken	transfer	11675	11505	0
RWAToken	transferFrom	11631	11478	0
[PASS] invariant_TotalSupplyConsistency() (runs: 256, calls: 128000, reverts: 104420)
Contract	Selector	Calls	Reverts	Discards
RWAToken	approve	11512	26	0
RWAToken	burn	11706	11557	0
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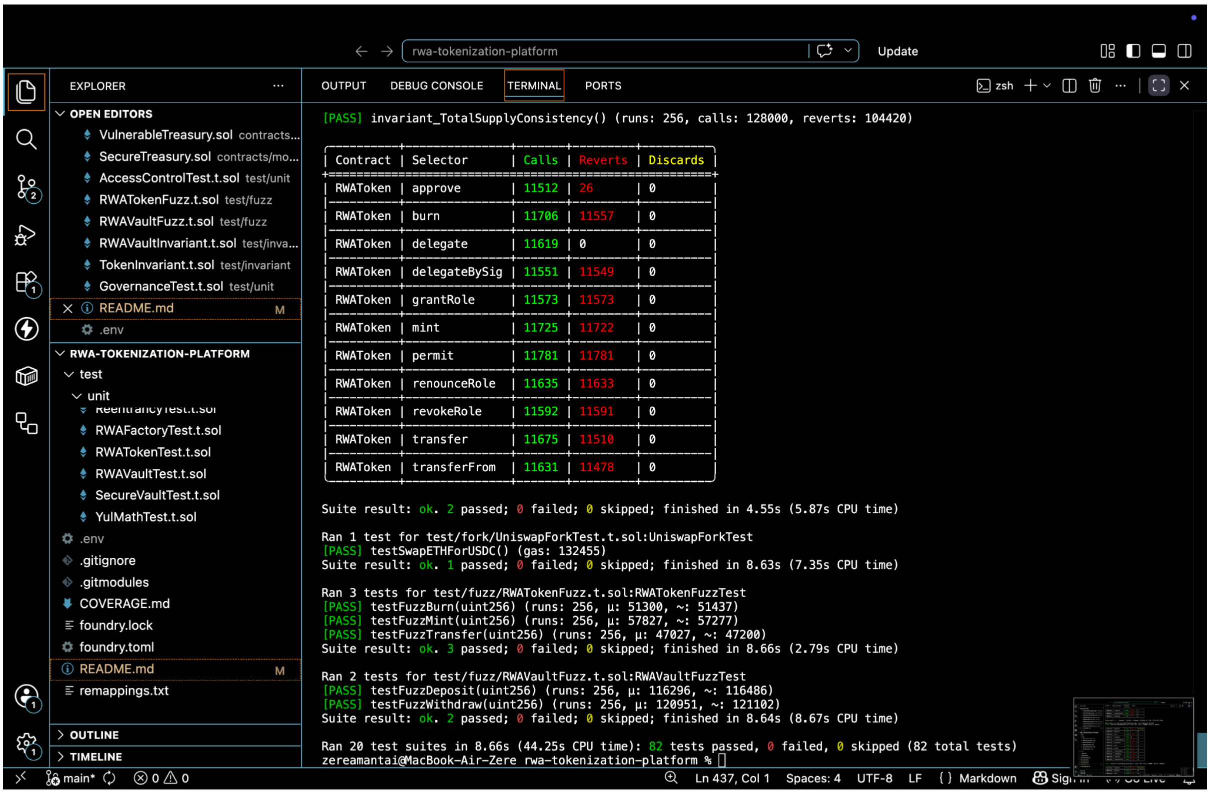
rwa-tokenization-platform
Update
EXPLORER
OPEN EDITORS
VulnerableTreasury.sol contracts...
SecureTreasury.sol contracts/mo...
AccessControlTest.t.sol test/unit
RWATokenFuzz.t.sol test/fuzz
RWAVaultFuzz.t.sol test/fuzz
RWAVaultInvariant.t.sol test/inva...
TokenInvariant.t.sol test/invariant
GovernanceTest.t.sol test/unit
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.env
RWA-TOKENIZATION-PLATFORM
test
unit
RWAFactoryTest.t.sol
RWATokenTest.t.sol
RWAVaultTest.t.sol
SecureVaultTest.t.sol
YulMathTest.t.sol
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.gitignore
.gitmodules
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foundry.lock
foundry.toml
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remappings.txt
OUTLINE
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OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
zsh
[PASS] invariant_TotalSupplyConsistency() (runs: 256, calls: 128000, reverts: 104420)
Contract	Selector	Calls	Reverts	Discards
RWAToken	approve	11512	26	0
RWAToken	burn	11706	11557	0
RWAToken	delegate	11619	0	0
RWAToken	delegateBySig	11551	11549	0
RWAToken	grantRole	11573	11573	0
RWAToken	mint	11725	11722	0
RWAToken	permit	11781	11781	0
RWAToken	renounceRole	11635	11633	0
RWAToken	revokeRole	11592	11591	0
RWAToken	transfer	11675	11510	0
RWAToken	transferFrom	11631	11478	0
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 4.55s (5.87s CPU time)
Ran 1 test for test/fork/UniswapForkTest.t.sol:UniswapForkTest
[PASS] testSwapETHForUSDC() (gas: 132455)
Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 8.63s (7.35s CPU time)
Ran 3 tests for test/fuzz/RWATokenFuzz.t.sol:RWATokenFuzzTest
[PASS] testFuzzBurn(uint256) (runs: 256, μ: 51300, ~: 51437)
[PASS] testFuzzMint(uint256) (runs: 256, μ: 57827, ~: 57277)
[PASS] testFuzzTransfer(uint256) (runs: 256, μ: 47027, ~: 47200)
Suite result: ok. 3 passed; 0 failed; 0 skipped; finished in 8.66s (2.79s CPU time)
Ran 2 tests for test/fuzz/RWAVaultFuzz.t.sol:RWAVaultFuzzTest
[PASS] testFuzzDeposit(uint256) (runs: 256, μ: 116296, ~: 116486)
[PASS] testFuzzWithdraw(uint256) (runs: 256, μ: 120951, ~: 121102)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 8.64s (8.67s CPU time)
Ran 20 test suites in 8.66s (44.25s CPU time): 82 tests passed, 0 failed, 0 skipped (82 total tests)
zereamantai@MacBook-Air-Zere rwa-tokenization-platform %
main*
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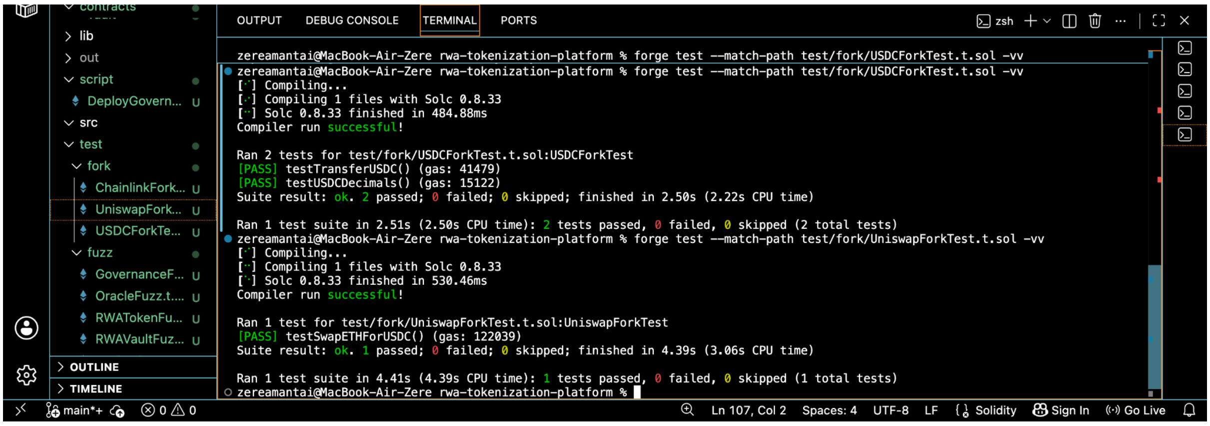
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
zereamantai@MacBook-Air-Zere rwa-tokenization-platform % forge test --match-path test/fork/USDCForkTest.t.sol -vv
zereamantai@MacBook-Air-Zere rwa-tokenization-platform % forge test --match-path test/fork/USDCForkTest.t.sol -vv
[⠊] Compiling...
[⠊] Compiling 1 files with Solc 0.8.33
[⠊] Solc 0.8.33 finished in 484.88ms
Compiler run successful!
Ran 2 tests for test/fork/USDCForkTest.t.sol:USDCForkTest
[PASS] testTransferUSDC() (gas: 41479)
[PASS] testUSDCDecimals() (gas: 15122)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 2.50s (2.22s CPU time)
Ran 1 test suite in 2.51s (2.50s CPU time): 2 tests passed, 0 failed, 0 skipped (2 total tests)
zereamantai@MacBook-Air-Zere rwa-tokenization-platform % forge test --match-path test/fork/UniswapForkTest.t.sol -vv
[⠊] Compiling...
[⠊] Compiling 1 files with Solc 0.8.33
[⠊] Solc 0.8.33 finished in 530.46ms
Compiler run successful!
Ran 1 test for test/fork/UniswapForkTest.t.sol:UniswapForkTest
[PASS] testSwapETHForUSDC() (gas: 122039)
Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 4.39s (3.06s CPU time)
Ran 1 test suite in 4.41s (4.39s CPU time): 1 tests passed, 0 failed, 0 skipped (1 total tests)
zereamantai@MacBook-Air-Zere rwa-tokenization-platform %
lib
out
script
DeployGovern...
src
test
fork
ChainlinkFork...
UniswapFork...
USDCForkTe...
fuzz
GovernanceF...
OracleFuzz.t....
RWATokenFu...
RWAVaultFuz...
OUTLINE
TIMELINE
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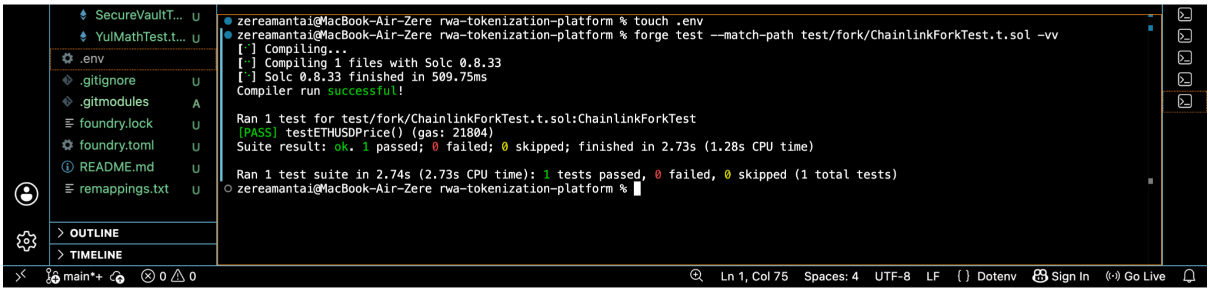
Solidity
Sign In
Go Live


SecureVaultT...
YulMathTest.t...
.env
.gitignore
.gitmodules
foundry.lock
foundry.toml
README.md
remappings.txt
OUTLINE
TIMELINE
zereamantai@MacBook-Air-Zere rwa-tokenization-platform % touch .env
zereamantai@MacBook-Air-Zere rwa-tokenization-platform % forge test --match-path test/fork/ChainlinkForkTest.t.sol -vv
[⠊] Compiling...
[⠊] Compiling 1 files with Solc 0.8.33
[⠊] Solc 0.8.33 finished in 509.75ms
Compiler run successful!
Ran 1 test for test/fork/ChainlinkForkTest.t.sol:ChainlinkForkTest
[PASS] testETHUSDPrice() (gas: 21804)
Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 2.73s (1.28s CPU time)
Ran 1 test suite in 2.74s (2.73s CPU time): 1 tests passed, 0 failed, 0 skipped (1 total tests)
zereamantai@MacBook-Air-Zere rwa-tokenization-platform %
main*+
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Spaces: 4
UTF-8
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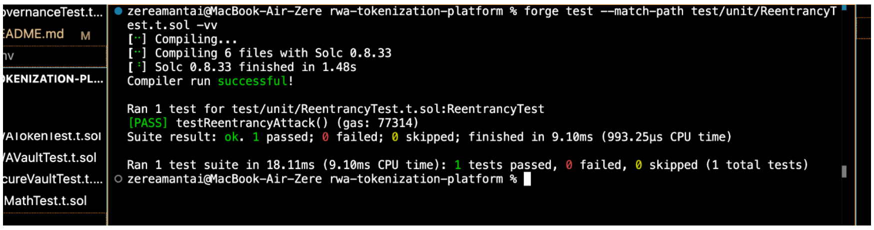
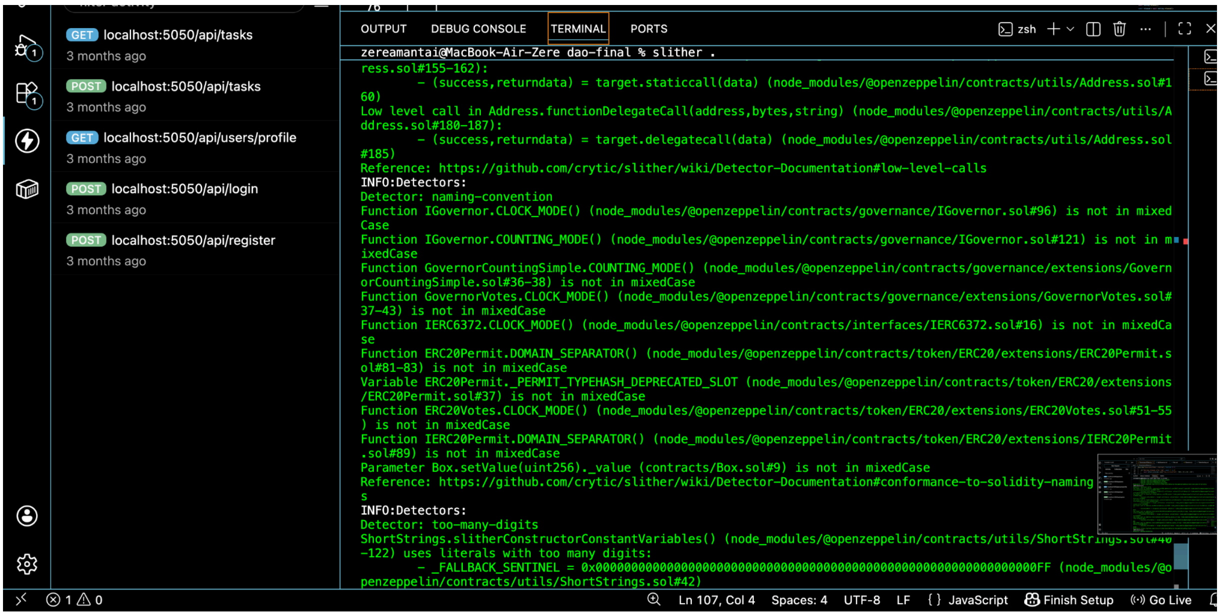
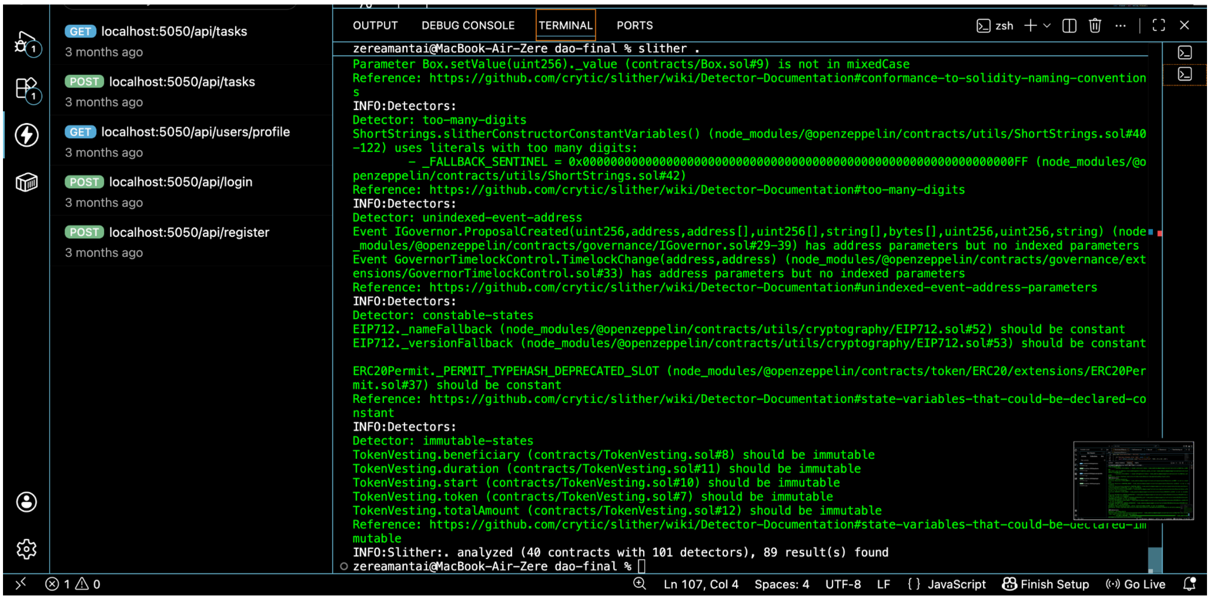
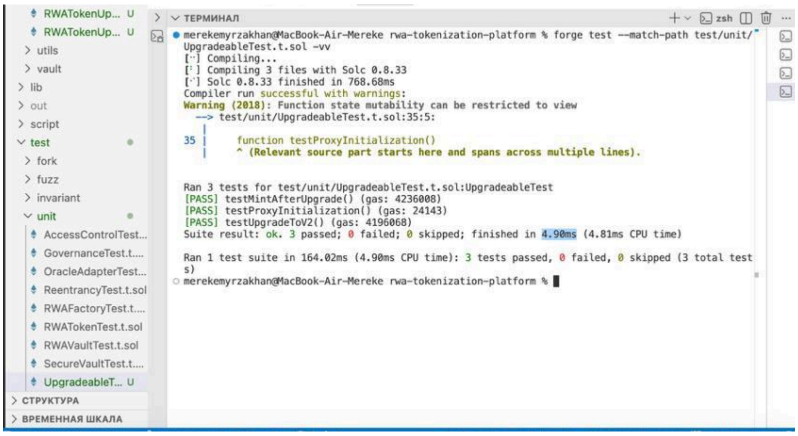
Go Live

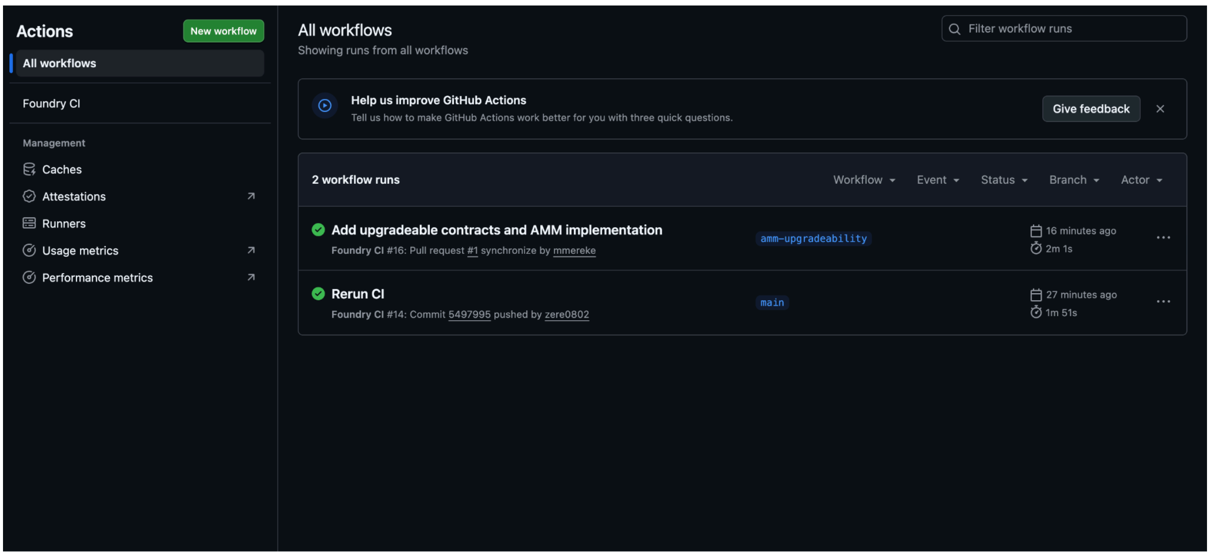

zereamantai@MacBook-Air-Zere rwa-tokenization-platform % forge test --match-path test/unit/ReentrancyTest.t.sol -vv
[⠊] Compiling...
[⠊] Compiling 6 files with Solc 0.8.33
[⠊] Solc 0.8.33 finished in 1.48s
Compiler run successful!
Ran 1 test for test/unit/ReentrancyTest.t.sol:ReentrancyTest
[PASS] testReentrancyAttack() (gas: 77314)
Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 9.10ms (993.25µs CPU time)
Ran 1 test suite in 18.11ms (9.10ms CPU time): 1 tests passed, 0 failed, 0 skipped (1 total tests)
zereamantai@MacBook-Air-Zere rwa-tokenization-platform %


GET localhost:5050/api/tasks
3 months ago
POST localhost:5050/api/tasks
3 months ago
GET localhost:5050/api/users/profile
3 months ago
POST localhost:5050/api/login
3 months ago
POST localhost:5050/api/register
3 months ago
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
zereamantai@MacBook-Air-Zere dao-final % slither .
ress.sol#155-162):
- (success,returndata) = target.staticcall(data) (node_modules/@openzeppelin/contracts/utils/Address.sol#160)
Low level call in Address.functionDelegateCall(address,bytes,string) (node_modules/@openzeppelin/contracts/utils/Address.sol#180-187):
- (success,returndata) = target.delegatecall(data) (node_modules/@openzeppelin/contracts/utils/Address.sol#185)
Reference: https://github.com/crytic/slither/wiki/Detector-Documentation#low-level-calls
INFO:Detectors:
Detector: naming-convention
Function IGovernor.CLOCK_MODE() (node_modules/@openzeppelin/contracts/governance/IGovernor.sol#96) is not in mixedCase
Function IGovernor.COUNTING_MODE() (node_modules/@openzeppelin/contracts/governance/IGovernor.sol#121) is not in mixedCase
Function GovernorCountingSimple.COUNTING_MODE() (node_modules/@openzeppelin/contracts/governance/extensions/GovernorCountingSimple.sol#36-38) is not in mixedCase
Function GovernorVotes.CLOCK_MODE() (node_modules/@openzeppelin/contracts/governance/extensions/GovernorVotes.sol#37-43) is not in mixedCase
Function IERC6372.CLOCK_MODE() (node_modules/@openzeppelin/contracts/interfaces/IERC6372.sol#16) is not in mixedCase
Function ERC20Permit.DOMAIN_SEPARATOR() (node_modules/@openzeppelin/contracts/token/ERC20/extensions/ERC20Permit.sol#81-83) is not in mixedCase
Variable ERC20Permit._PERMIT_TYPEHASH_DEPRECATED_SLOT (node_modules/@openzeppelin/contracts/token/ERC20/extensions/ERC20Permit.sol#37) is not in mixedCase
Function ERC20Votes.CLOCK_MODE() (node_modules/@openzeppelin/contracts/token/ERC20/extensions/ERC20Votes.sol#51-55) is not in mixedCase
Function IERC20Permit.DOMAIN_SEPARATOR() (node_modules/@openzeppelin/contracts/token/ERC20/extensions/IERC20Permit.sol#89) is not in mixedCase
Parameter Box.setValue(uint256)._value (contracts/Box.sol#9) is not in mixedCase
Reference: https://github.com/crytic/slither/wiki/Detector-Documentation#conformance-to-solidity-naming
s
INFO:Detectors:
Detector: too-many-digits
ShortStrings.slitherConstructorConstantVariables() (node_modules/@openzeppelin/contracts/utils/ShortStrings.sol#40-122) uses literals with too many digits:
- _FALLBACK_SENTINEL = 0x00000000000000000000000000000000000000000000000000000000000000FF (node_modules/@openzeppelin/contracts/utils/ShortStrings.sol#42)
Ln 107, Col 4
Spaces: 4
UTF-8
LF
JavaScript
Finish Setup
Go Live


GET localhost:5050/api/tasks
3 months ago
POST localhost:5050/api/tasks
3 months ago
GET localhost:5050/api/users/profile
3 months ago
POST localhost:5050/api/login
3 months ago
POST localhost:5050/api/register
3 months ago
OUTPUT DEBUG CONSOLE TERMINAL PORTS
zereamantai@MacBook-Air-Zere dao-final % slither .
Parameter Box.setValue(uint256)._value (contracts/Box.sol#9) is not in mixedCase
Reference: https://github.com/crytic/slither/wiki/Detector-Documentation#conformance-to-solidity-naming-conventions
INFO:Detectors:
Detector: too-many-digits
ShortStrings.slitherConstructorConstantVariables() (node_modules/@openzeppelin/contracts/utils/ShortStrings.sol#40-122) uses literals with too many digits:
- _FALLBACK_SENTINEL = 0x00000000000000000000000000000000000000000000000000000000000000FF (node_modules/@openzeppelin/contracts/utils/ShortStrings.sol#42)
Reference: https://github.com/crytic/slither/wiki/Detector-Documentation#too-many-digits
INFO:Detectors:
Detector: unindexed-event-address
Event IGovernor.ProposalCreated(uint256,address,address[],uint256[],string[],bytes[],uint256,uint256,string) (node_modules/@openzeppelin/contracts/governance/IGovernor.sol#29-39) has address parameters but no indexed parameters
Event GovernorTimelockControl.TimelockChange(address,address) (node_modules/@openzeppelin/contracts/governance/extensions/GovernorTimelockControl.sol#33) has address parameters but no indexed parameters
Reference: https://github.com/crytic/slither/wiki/Detector-Documentation#unindexed-event-address-parameters
INFO:Detectors:
Detector: constable-states
EIP712._nameFallback (node_modules/@openzeppelin/contracts/utils/cryptography/EIP712.sol#52) should be constant
EIP712._versionFallback (node_modules/@openzeppelin/contracts/utils/cryptography/EIP712.sol#53) should be constant
ERC20Permit._PERMIT_TYPEHASH_DEPRECATED_SLOT (node_modules/@openzeppelin/contracts/token/ERC20/extensions/ERC20Permit.sol#37) should be constant
Reference: https://github.com/crytic/slither/wiki/Detector-Documentation#state-variables-that-could-be-declared-constant
INFO:Detectors:
Detector: immutable-states
TokenVesting.beneficiary (contracts/TokenVesting.sol#8) should be immutable
TokenVesting.duration (contracts/TokenVesting.sol#11) should be immutable
TokenVesting.start (contracts/TokenVesting.sol#10) should be immutable
TokenVesting.token (contracts/TokenVesting.sol#7) should be immutable
TokenVesting.totalAmount (contracts/TokenVesting.sol#12) should be immutable
Reference: https://github.com/crytic/slither/wiki/Detector-Documentation#state-variables-that-could-be-declared-immutable
INFO:Slither:. analyzed (40 contracts with 101 detectors), 89 result(s) found
zereamantai@MacBook-Air-Zere dao-final %
Ln 107, Col 4 Spaces: 4 UTF-8 LF { } JavaScript Finish Setup Go Live


RWATokenUp... U
RWATokenUp... U
utils
vault
lib
out
script
test
fork
fuzz
invariant
unit
AccessControlTest...
GovernanceTest.t....
OracleAdapterTest...
ReentrancyTest.t.sol
RWAFactoryTest.t....
RWATokenTest.t.sol
RWAVaultTest.t.sol
SecureVaultTest.t....
UpgradeableT... U
СТРУКТУРА
ВРЕМЕННАЯ ШКАЛА
ТЕРМИНАЛ
merekemyrzakhan@MacBook-Air-Mereke rwa-tokenization-platform % forge test --match-path test/unit/UpgradeableTest.t.sol -vv
[⠊] Compiling...
[⠊] Compiling 3 files with Solc 0.8.33
[⠊] Solc 0.8.33 finished in 768.68ms
Compiler run successful with warnings:
Warning (2018): Function state mutability can be restricted to view
--> test/unit/UpgradeableTest.t.sol:35:5:
35 | function testProxyInitialization()
^ (Relevant source part starts here and spans across multiple lines).
Ran 3 tests for test/unit/UpgradeableTest.t.sol:UpgradeableTest
[PASS] testMintAfterUpgrade() (gas: 4236008)
[PASS] testProxyInitialization() (gas: 24143)
[PASS] testUpgradeToV2() (gas: 4196068)
Suite result: ok. 3 passed; 0 failed; 0 skipped; finished in 4.90ms (4.81ms CPU time)
Ran 1 test suite in 164.02ms (4.90ms CPU time): 3 tests passed, 0 failed, 0 skipped (3 total tests)
merekemyrzakhan@MacBook-Air-Mereke rwa-tokenization-platform %


AccessControlTest...
GovernanceTest.t....
OracleAdapterTest...
ReentrancyTest.t.sol
RWAFactoryTest.t....
RWATokenTest.t.sol
RWAVaultTest.t.sol
SecureVaultTest.t....
SimpleAMMTest.t....
СТРУКТУРА
ВРЕМЕННАЯ ШКАЛА
merekemyrzakhan@MacBook-Air-Mereke rwa-tokenization-platform % forge test --match-path test/unit/SimpleAMMTest.t.sol -vv
[⠊] Compiling...
[⠊] Compiling 1 files with Solc 0.8.33
[⠊] Solc 0.8.33 finished in 723.98ms
Compiler run successful!
Ran 5 tests for test/unit/SimpleAMMTest.t.sol:SimpleAMMTest
[PASS] testAddLiquidity() (gas: 234878)
[PASS] testKInvariant() (gas: 249437)
[PASS] testRemoveLiquidity() (gas: 196619)
[PASS] testSwapAForB() (gas: 252022)
[PASS] testSwapBForA() (gas: 252233)
Suite result: ok. 5 passed; 0 failed; 0 skipped; finished in 5.72ms (6.70ms CPU time)
Ran 1 test suite in 172.56ms (5.72ms CPU time): 5 tests passed, 0 failed, 0 skipped (5 total tests)


Actions
New workflow
All workflows
Foundry CI
Management
Caches
Attestations
Runners
Usage metrics
Performance metrics
All workflows
Showing runs from all workflows
Filter workflow runs
Help us improve GitHub Actions
Tell us how to make GitHub Actions work better for you with three quick questions.
Give feedback
2 workflow runs
Workflow Event Status Branch Actor
Add upgradeable contracts and AMM implementation
Foundry CI #16: Pull request #1 synchronize by mmereke
amm-upgradeability
16 minutes ago
2m 1s
Rerun CI
Foundry CI #14: Commit 5497995 pushed by zere0802
main
27 minutes ago
1m 51s